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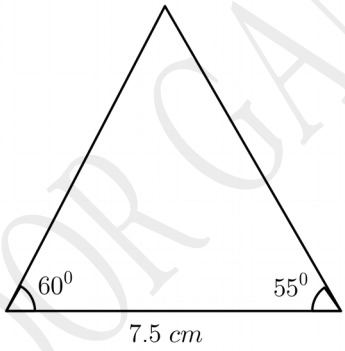
A

SUMMATIVE ASSESSMENT- II 2025 – 26

Class : IX

MATHEMETICS – ANSWER KEY

E- 903

Qn no	Key	Score
	SECTION A	
1	C. 10	1
2	B. $2 + -5 = -3$	1
3	B . 8 cm^2	1
4	C. $\sqrt{2}$, $\sqrt{8}$	1
5	C. $\sqrt{5}$	1
6	C. Both (ii) and (iv) are true	1
7	C. Both statements are true , Statement 2 is the reason for Statement 1.	1
8	D. Both statements are true , Statement 2 is not the reason for Statement 1.	1
	SECTION B	
9	For drawing a line of length 7.5 cm For drawing 60^0 and 55^0 angles on the both ends of this line and completing the triangle.  	1 2
10	(i) $a - b = -5 - 4 = -9$ (ii) $b - c = 4 - (-3) = 4 + 3 = 7$ (iii) $a + b - c = -5 + 7 = 7 - 5 = 2$	1 1 1
11A	$\frac{\sqrt{108}}{3}$ $= \frac{\sqrt{6 \times 6 \times 3}}{3} = \frac{6\sqrt{3}}{3} = 2\sqrt{3}$ $= 2 \times 1.73 = 3.46\ cm = 34.6\ mm$	1 1 1
11B	(i) $(2 + \sqrt{3})(2 - \sqrt{3}) = 2^2 - (\sqrt{3})^2 = 4 - 3 = 1$ (ii) $\frac{1}{2 - \sqrt{3}} = 2 + \sqrt{3}$ $= 2 + 1.73 = 3.73$	1 1 1
12	(i) $\lvert x \rvert + \lvert y \rvert = \lvert 4 \rvert + \lvert -3 \rvert = 4 + 3 = 7$ (ii) $\lvert x + y \rvert = \lvert 4 + (-3) \rvert = \lvert 1 \rvert = 1$ (iii) $x = 4$, $y = 3$	1 1 1

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13A

Sum of the lengths of the sides $= \frac{46}{2} = 23\ cm$ — 1

Difference of the lengths of the sides $= 7\ cm$

Length of the longer side $= \frac{23 + 7}{2} = \frac{30}{2} = 15\ cm$

Length of the shorter side $= \frac{23 - 7}{2} = \frac{16}{2} = 8\ cm$ — 1

OR — 1

If we take the lengths of the sides as $x\ cm$ and $y\ cm$, then

$2x + 2y = 46 \implies x + y = \frac{46}{2} = 23$

$x - y = 7$

$x = \frac{23 + 7}{2} = \frac{30}{2} = 15\ cm \quad , \quad y = \frac{23 - 7}{2} = \frac{16}{2} = 8\ cm$

13B

If the excess 550 Rs is not taken ,

the total prices of a boot and two jerseys $= 1600 - 550 = 1050\ Rs$ — 1

$\therefore$ The price of a jersey $= \frac{1050}{3} = 350\ Rs$ — 1

The price of a boot $= 350 + 550 = 900\ Rs$ — 1

OR

If we take the price of a boot $= x\ Rs$ and that of a jersey $= y\ Rs$, then

$x + 2y = 1600 \quad (1)$

$x = y + 550 \quad (2)$

Put $x = y + 550$ in equation (1), we get

$y + 550 + 2y = 1600$

$3y + 550 = 1600 \implies 3y = 1600 - 550 = 1050 \implies y = \frac{1050}{3} = 350\ Rs$

From equation (1), $x = 350 + 550 = 900\ Rs$

14

$p(x) = 5x^2 + 3x + 1$

$p(1) = 5 + 3 + 1 = 9$ — 1

$p(0) = 1$ — 1

$p(-2) = 5 \times (-2)^2 + 3 \times (-2) + 1 = 5 \times 4 - 6 + 1 = 20 - 6 + 1 = 15$ — 1

SECTION C

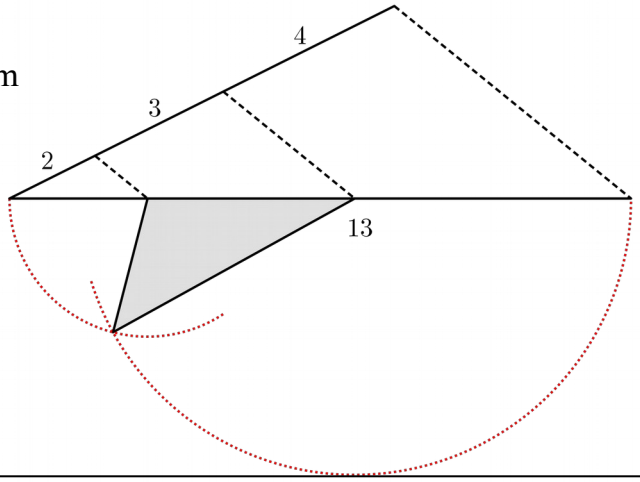
15

For drawing a line of length 13 cm. — 1

For drawing a slanted line of length 9 cm at one end of this line . — 1

For dividing the 13 cm long line in the ratio 2 : 3 : 4 — 1

For completing the triangle. — 1



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16	Radius of the circle = Length of a side of the regular hexagon = 3 cm (i) Perimeter of the regular hexagon $= 6 \times 3 = 18\ cm$ (ii) Circumference of the circle $= 2\pi \times 3 = 6\pi = 6 \times 3.14 = 18.84\ cm$ (iii) Difference between the perimeters $= 18.84 - 18 = 0.84\ cm$ $= 0.84 \times 10 = 8.4\ cm$	1 1 1 1
17A	(i) If we take in triangle PQR, $\angle Q = x^0$, then $\angle R = 90 - x^0$ $[\ \angle QPR = 90^0\ ]$ $\therefore\ \angle QPS = 90 - x^0$ $[\ \angle QSP = \angle PSR = 90^0\ ]$ $\angle SPR = x^0$ (ii) $\frac{h}{b} = \frac{a}{h}$ $\left[\ \frac{PS}{SR} = \frac{QS}{PS} = \frac{PQ}{PR}\ \right]$ $h \times h = a \times b \quad ==> \quad h^2 = ab$	1 1 1 1 1
17B	$\angle BAP = \angle CAP = x^0$ [AP is the bisector of $\angle$ A] $\angle QAC = \angle AQC = x^0$ [AC = QC] $\angle APB = \angle CPQ = y^0$ [Opposite angles are equal] (i) $\angle BAP = \angle CQP = x^0$ $\angle APB = \angle CPQ = y^0$ $\therefore\ \angle ABP = \angle PCQ$ (ii) $\frac{PB}{PC} = \frac{AB}{QC}$ $\left[\ \frac{PB}{PC} = \frac{AB}{QC} = \frac{AP}{PQ}\ \right]$ $\frac{PB}{PC} = \frac{AB}{AC}$	1 1 1 1
18	(i) $y = (-1-1)^2 = (-2)^2 = 4$ (ii) $y = 3^2 - 6 \times 3 + 9 = 9 - 18 + 9 = 0$ $y = (-1)^2 - 6 \times (-1) + 9 = 1 + 6 + 9 = 16$ y can not be negative. $y = x^2 - 6x + 9 = x^2 - 2 \times x \times 3 + 3^2 = (x-3)^2$ The square of any number is always non negative.	1 1 1 1
19	(i) Perimeter of the triangle $= 9 + 10 + 17 = 36$ (ii) $s = \frac{36}{2} = 18$ Area $= \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{18\,(18-9)(18-10)(18-17)}$ $= \sqrt{18 \times 9 \times 8 \times 1}$ $= \sqrt{(2 \times 3 \times 3) \times (3 \times 3) \times (2 \times 2 \times 2)}$ $= 2 \times 3 \times 3 \times 2 = 36\ sq.cm$	1 1 1 1 1
20A	(i) $\lvert x-2 \rvert = 3$ $x = 2 + 3 = 5$ OR $x = 2 - 3 = -1$ (ii) $\lvert x+4 \rvert = \lvert x-6 \rvert$ $\lvert x-(-4) \rvert = \lvert x-6 \rvert$ $x = \frac{-4+6}{2} = \frac{6-4}{2} = \frac{2}{2} = 1$	2 1 1

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20B

(i) $4\ ,\ -4$ — 2

(ii) Diameter $= |-3-7| = |-10| = 10$ — 1

Number denoting the centre of the circle $= \dfrac{-3+7}{2} = \dfrac{7-3}{2} = \dfrac{4}{2} = 2$ — 1

SECTION D

21

By changing the sides and the diagonal of the given quadrilateral by $1\frac{1}{2}$ times. — 3

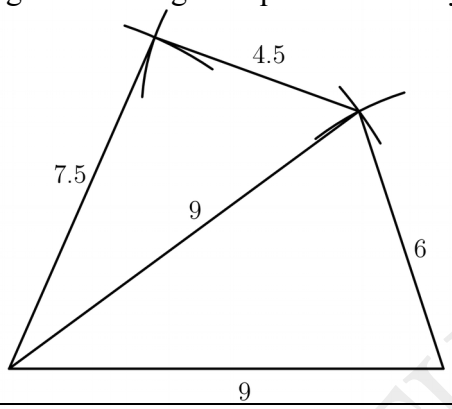
For completing the quadrilateral. — 2

$6 \times 1\frac{1}{2} = 6 + 3 = 9$

$4 \times 1\frac{1}{2} = 4 + 2 = 6$

$5 \times 1\frac{1}{2} = 5 + 2\frac{1}{2} = 7\frac{1}{2}$

$3 \times 1\frac{1}{2} = 3 + 1\frac{1}{2} = 4\frac{1}{2}$]



22

(i) $AB = \sqrt{32} = \sqrt{16 \times 2} = 4\sqrt{2}\ cm$ — 1

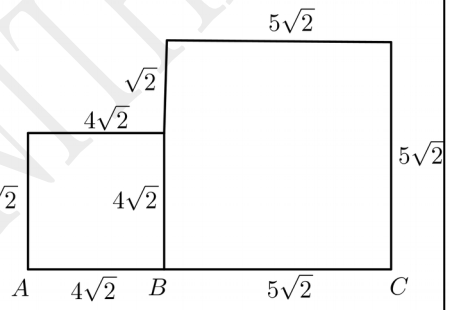
(ii) $BC = \sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2}\ cm$ — 1

$AC = 4\sqrt{2} + 5\sqrt{2} = 9\sqrt{2}\ cm$ — 1

(iii) Perimeter $= 9\sqrt{2} + 5\sqrt{2} + 5\sqrt{2} + \sqrt{2} + 4\sqrt{2} + 4\sqrt{2}$

$= 28\sqrt{2} = 28 \times 1.41 = 39.48\ cm$ — 1

$= 39.48 \times 10 = 394.8\ mm$ — 1



23A

(i) $a = (4+3) \times -5 = 7 \times -5 = -35$ — 1

$b = (4-3) \times -5 = 1 \times -5 = -5$ — 1

$c = (-6+(-1)) \times 2 = -\ -7 \times 2 = -14$ — 1

$d = (-6-(-1)) \times 2 = -\ (-6 \pm 1) \times 2 = -5 \times 2 = -10$ — 1

(ii) $x = 1\ ,\ \ y = -1$ $\quad [\ 1^2 - (-1)^2 = 1 - 1 = 0\ ]$ — 1

23B

(i) $y = \dfrac{1}{x} = \dfrac{3}{-1} = -3$ — 1

(ii) $y = \dfrac{1}{-3+1} + \dfrac{1}{-3-1} = \dfrac{1}{1-3} + \dfrac{1}{-4} = \dfrac{1}{-2} + \dfrac{1}{-4} = \dfrac{-1}{2} + \dfrac{-1}{4} = \dfrac{-3}{4}$ — 2

(iii) $y = \dfrac{x}{y} - \dfrac{y}{x} = \dfrac{-4}{-2} - \dfrac{-2}{-4} = 2 - \dfrac{1}{2} = 1\dfrac{1}{2}$ — 2

24A

If we take the lengths of the sides of the rectangle as $x\ m$ and $y\ m$, then

$(x-1)(y-1) = 1400 \implies xy - (x+y) + 1 = 1400$

$\implies xy - (x+y) = 1400 - 1 = 1399 \quad (1)$ — 1

$(x+1)(y+1) = 1554 \implies xy + (x+y) + 1 = 1554$

$\implies xy + (x+y) = 1554 - 1 = 1553 \quad (2)$ — 1

Adding equations (1) and (2),

$2xy = 2952 \implies xy = \dfrac{2952}{2} = 1476$ — 1

(i) Area of the rectangle $= 1476\ sq.m$

Put $xy = 1476$ in equation (2),

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	$x + y = 1553 - 1476 = 77$ Perimeter of the rectangle $= 2(x + y) = 2 \times 77 = 154\ m$	1 1
24B	(i) $16 \times 4 - (2 \times 18) = 64 - 36 = 28$ (ii) If we take the smallest number as x , then the numbers are $x\ ,\ x + 2\ ,\ x + 14\ ,\ x + 16$ Difference of the diagonal products $= (x + 14)(x + 2) - x(x + 16)$ $= x^2 + (14 + 2)x + 28 - [x^2 + 16x]$ $= x^2 + 16x + 28 - x^2 - 16x$ $= 28$	1 1 1 1 1
25A	(i) Area of the circle $= \pi r^2 = \pi \times 6^2 = 36\pi\ sq.cm$ (ii) Area of the triangle $= \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4} \times 6^2 = 9\sqrt{3}\ sq.cm$ (iii) Area of the shaded part = Area of the circle − Area of the triangle $= 36\pi - 9\sqrt{3}$ $= 36 \times 3.14 - (9 \times 1.73)$ $= 113.04 - 15.57$ $= 97.47\ sq.cm$	1 1 1 1 1
25B	(i) Radius of the semicircle = 2 cm (ii) Area of the rectangle $= 8 \times 4 = 32\ sq.cm$ (iii) Area of the shaded part = Area of the rectangle − 2 × Area of the the circle $= 32 - 2 \times \pi \times 2^2$ $= 32 - 8\pi$ $= 32 - (8 \times 3.14)$ $= 32 - 25.12 = 6.88\ sq.cm$ OR Area of the shaded part = Area of the rectangle − 4 × Area of the the semicircle $= 32 - 4 \times \frac{1}{2} \times \pi \times 2^2$ $= 32 - 8\pi$ $= 32 - (8 \times 3.14)$ $= 32 - 25.12 = 6.88\ sq.cm$	1 1 1 1 1
26	(i) $2 + x\ ,\ 3 + x\ ,\ 5 + x$ (ii) $v(x) = (2 + x)(3 + x)(5 + x) = x^3 + 10x^2 + 31x + 30$ (iii) $v(1) = 1 + 10 + 31 + 30 = 72$ OR $v(1) = (2 + 1)(3 + 1)(5 + 1) = 3 \times 4 \times 6 = 72$ `	2 2 1

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